

Pon Aung

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Portfolio: PonAppliedPhysics.github.io

EDUCATION

The Ohio State University

Bachelor of Science in Physics

Minor: Engineering Science | Mechanical Engineering Emphasis

Columbus, OH

May 2026

EXPERIENCE

Honda Manufacturing of America

Project Leader – BEV Thermal Runaway Detection Capstone

Marysville, OH

August 2024 – May 2025

- Narrowed 9 thermal runaway detection technologies to 4 sensor candidates (pyrometer, MEMS thermal, HF gas, ultrasonic) through weighted decision analysis against a stakeholder-driven safety hierarchy for Honda's 800-lb BEV battery packs
- Defined design requirements through on-site stakeholder interviews with Honda safety and BEV engineers, establishing that the detection system must operate independently of assembly line controls a constraint that eliminated 5 of 9 candidate technologies before prototyping
- Delivered sensor validation data, selection matrix, and deployment recommendations to Honda engineering leadership, completing all laboratory testing at \$1,268 — 36% of the \$3,500 budget and advancing the project to prototype phase

PROJECT EXPERIENCE

OSU Underwater Robotics — Talos AUV

Mechanical Team Member

Columbus, OH

December 2025 – Present

- Designed hull and chassis components in SolidWorks and machined parts on lathe, mill, and 3D printer to integrate a dual stereo-camera vision system (ZED X front-facing, ZED X Mini downward-facing) into the Talos AUV for the RoboSub competition
- Redesigned torpedo launcher servo housings in Delrin with built-in mechanical seals and a custom gearbox to convert 270-degree servo rotation to full range of motion, eliminating all water-ingress failures from the previous RoboSub 2024 design
- Integrated mechanical, electrical, and software subsystems through iterative pool testing, adding a structural brace and guiding rod to the torpedo interface that improved launch accuracy over the prior competition design

Computational Heat Transfer Analysis and Experimental Validation

Advanced Physics [github](#)

Columbus, OH

Spring 2026

- Validated 11-thermocouple sensor array data by measuring and subtracting per-sensor systematic offsets ($\sim 0.6^{\circ}\text{C}$), then confirming cross-run consistency between two independent cooling experiments
- Developed 5 progressively refined finite-difference models, reducing chi-squared error by 10x+ through data-driven boundary conditions and SciPy-optimized radial heat loss; vectorized NumPy implementation cut runtimes from minutes to 3 seconds
- Built a physics-informed neural network (PINN) in TensorFlow that recovered thermal diffusivity as a trainable parameter directly from sensor data, achieving MAE of 0.42°C and RMSE of 0.53°C without manual parameter tuning
- Identified and corrected dimensional errors in published reference materials (rod length, thermocouple convention), preventing systematic bias across all five computational models

WORK EXPERIENCE

Barcelona

Server

Columbus, OH

February 2023 – August 2025

- Managed high-volume table sections nightly in an upscale Spanish restaurant, maintaining detailed knowledge of rotating wine lists and seasonal tapas menus to drive guest recommendations

TECHNICAL SKILLS

Software and Tools: MATLAB, Python (NumPy, SciPy, Pandas, TensorFlow, Scikit-learn), SolidWorks, Shapr3D, LabPlot

Computational Methods: Finite-difference PDE solvers, parameter optimization, stability analysis, chi-squared goodness-of-fit testing, time-series analysis, weighted least-squares regression, uncertainty propagation

Machine Learning: Supervised learning (regression, random forests), physics-informed neural networks (PINNs), model validation (RMSE, MAE, R^2)

Mechanical Engineering: Heat Transfer, Statics, Dynamics, Structural Analysis, Circuits, Lab Instrumentation and Data Collection

ACTIVITIES

Underwater Robotics Club, The Ohio State University

December 2025 – May 2026

American Society of Mechanical Engineers (ASME), The Ohio State University

December 2025 – May 2026